

Generative Classifiers

Types of Classifiers

- Generative: learns to generate the examples
 - Learn $P(x|y)$ and $P(y)$ from the training data and then apply Bayes rule to find $P(y|x)$

$$\underline{P(y|x)} = \frac{P_{\theta}(x|y) P(y)}{\sum_{y'} \underbrace{P_{\theta}(x|y') P(y')}_{P(x)}}$$

Types of Classifiers

or discriminative

- Conditional Classifiers: Model conditional distribution $P(\underline{y}|x)$ directly.
 - Example: logistic regression classifier
 - Neural Networks.

Generative classifiers

- Modeling $P(y)$: Easy, k-way multinomial distribution for k-classes.
- Modeling $P(x|y)$: Challenges, x is high-dimensional making estimating this probability challenging
 $x \in \mathbb{R}^d$
- Spaces required:

$$P(x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} | y) =$$

- 1) You can make simplifying assumption about form of $P(x|y)$ (eg Gaussian) but that may introduce mismatch with true data distribution
- 2) More powerful estimators like kernel density could make inference expensive -

Two Generative Models with simplifying assumptions

- Naïve Bayes classifier
- Gaussian Discriminate analysis.

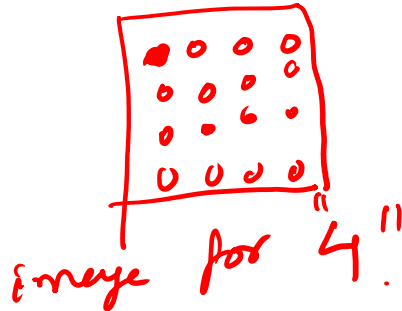
Naive Bayes Classifier: A Generative Classifier

Each attribute $\underline{x_j}$ is conditionally independent given the class label

Formula: $P(\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} | y) = \prod_{j=1}^d P(\underline{x_j} | y)$ ✓

$$P(\underline{x}) = \prod_{j=1}^d P(x_j)$$

↙ does not imply



Conditional Independence for documents

$y = \{\text{"garbage", "food"}\}$

$\{-\text{asty}, \text{stale}, \text{bland}, \text{trash}, \text{collect}\}$

$x: x_1, x_2$ $p(x_1 = \text{taste}, x_2 = \text{bland})$ $\checkmark$ Correlated
"food" $x_1 \perp x_2$
 $p(x_1 = \text{taste} \mid y = \text{food}) \perp p(x_2 = \text{bland} \mid y = \text{food})$

Training a Naïve Bayesian Classifier

- Given training data, apply maximum likelihood principle to estimate parameters $\underline{P(y)} ; \underline{P(x_j|y)}$ for each $j=1 \dots d$ $Kd + 1$
- Estimating $\underline{P(y)}$: A multinomial distribution.
- Estimating $\underline{P(x_j|y)}$
 - If j-th attribute is categorical: $\underline{P(x_j|y)}$ is estimated as the relative freq of samples having value d_i as j-th attribute in class y
 - If j-th attribute is continuous: $\underline{P(x_j|y)}$ is estimated through a continuous density function: eg. Gaussian density function
- Computationally easy in both cases

Play-tennis example: estimating $P(x_i|C)$

D

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

$P(y)$ $y \in \{P, N\}$

$P(p) = 9/14$
$P(n) = 5/14$

outlook	$P(x_1 y)$ $y=p$ $y=n$
$P(\text{sunny} p) = 2/9$	$P(\text{sunny} n) = 3/5$
$P(\text{overcast} p) = 4/9$	$P(\text{overcast} n) = 0$
$P(\text{rain} p) = 3/9$	$P(\text{rain} n) = 2/5$
temperature	$P(x_2 y)$
$P(\text{hot} p) = 2/9$	$P(\text{hot} n) = 2/5$
$P(\text{mild} p) = 4/9$	$P(\text{mild} n) = 2/5$
$P(\text{cool} p) = 3/9$	$P(\text{cool} n) = 1/5$
humidity	$P(x_3 y)$
$P(\text{high} p) = 3/9$	$P(\text{high} n) = 4/5$
$P(\text{normal} p) = 6/9$	$P(\text{normal} n) = 2/5$
windy	$P(x_4 y)$
$P(\text{true} p) = 3/9$	$P(\text{true} n) = 3/5$
$P(\text{false} p) = 6/9$	$P(\text{false} n) = 2/5$

Naive Bayesian Classifier (II)

- Given a training set, we can compute the probabilities

Outlook	P	N	Humidity	P	N
sunny	2/9	3/5	high	3/9	4/5
overcast	4/9	0	normal	6/9	1/5
rain	3/9	2/5			
Temperature			Windy		
hot	2/9	2/5	true	3/9	3/5
mild	4/9	2/5	false	6/9	2/5
cool	3/9	1/5			

0 probabilities are a problem.

In practice we smooth estimates

eg: Lidstone or

Laplace smoothing

$$P(x_2 = \text{cool} | y = P)$$

$$P(x_j = v | y) = \frac{n_{jv|y} + 1}{\sum_{v_i} n_{jv_i|y} + m}$$

$m \equiv \text{Domain of } x_j$

0

Play-tennis example: classifying X

- An unseen sample $X = \langle \text{rain, hot, high, false} \rangle$
- $P(X|p) \cdot P(p) =$
 $P(\text{rain}|p) \cdot P(\text{hot}|p) \cdot P(\text{high}|p) \cdot P(\text{false}|p) \cdot P(p)$
 - $= 3/9 \cdot 2/9 \cdot 3/9 \cdot 6/9 \cdot 9/14 = 0.010582$
- $P(X|n) \cdot P(n) = P(\text{rain}|n) \cdot P(\text{hot}|n) \cdot P(\text{high}|n) \cdot P(\text{false}|n) \cdot P(n)$
 - $= 2/5 \cdot 2/5 \cdot 4/5 \cdot 2/5 \cdot 5/14 = 0.018286$
- Sample X is classified in class n (don't play)

$$P(y = p | X) = \frac{0.010582}{0.010582 + 0.018286} < \frac{1}{2}$$

Naive Bayes Spam Filter

- Data
 - Emails (headers, body, metadata)
 - Labels (spam/ham)
assume that users actually label all mails
 - Images and labels
- Need to estimate $p(y)$, $p(x_i|y)$
 - Compute distribution of x_i for every y
 - Compute distribution of y

Text
vocabulary of words V
 $|V| = d$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

$x_j = 1$ iff j -th word
of vocabulary is present
in document x

$$P(x_j|y) \sim \text{Bernoulli}(p_{j,y})$$

Naive NaiveBayes Classifier

- Two classes (spam/ham)
- Binary features (e.g. presence of \$\$\$, viagra)
- Simplistic Algorithm
 - Count occurrences of feature for spam/ham
 - Count number of spam/ham mails

Training data
 $n = \#$ of documents
 $n(y) = \#$ of label y
 $n(i, y) = \#$ of document with label y and word i

feature probability

$$p(x_i = \text{TRUE} | y) = \frac{n(i, y)}{n(y)} \text{ and } p(y) = \frac{n(y)}{n}$$

spam probability

$p(y)$

$$p(y|x) \propto \frac{n(y)}{n} \prod_{i:x_i=\text{TRUE}} \frac{n(i, y)}{n(y)} \prod_{i:x_i=\text{FALSE}} \left(\frac{n(y) - n(i, y)}{n(y)} \right)$$

Summary of Naïve Bayes Classifier

- Creates a distribution over class label given x $P(y|x)$ by applying Bayes rule.
 - Requires estimating $P(x|y)$ for each class y and $P(y)$
- Estimates $P(x|y)$ by assuming that each attributes of x are conditionally independent given the class label
 - Very easy computationally. $P(x_i|y)$
- Many applications in spite of simplistic assumption: e.g. classifying emails as spam Vs non-spam
- Limitations of generative method: estimating $P(x|y)$ is hard since x could be high-dimensional. Useful when $P(x|y)$ is already available, e.g. in speech recognition use of HMMs for word recognition } ignore this

Demo

https://colab.research.google.com/drive/1_9j1CAvkPHq18zhZ3tBj4r3l7-LZKZ3c?usp=sharing

Gaussian Discriminate Analysis

Assume that $P(x|y) \sim N(x; \mu_y; \Sigma_y)$
 $x \in \mathbb{R}^d, \mu \in \mathbb{R}^d \quad \Sigma \sim \mathbb{R}^{d \times d} \quad -(x - \mu_y)^T \Sigma_y^{-1} (x - \mu_y)$

Two types of approaches

- 1) Each class label has its own co-variance parameters (Quadratic Discriminate Analysis)
- 2) Classes share a co-variance matrix.
(Linear Discriminate Analysis).

Assume $k=2$

Analyze the decision boundary for the GDA methods.

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

characterize all x s.t. $\underline{P(y=1|x)} > \underline{P(y=2|x)}$

$$P(x|y=1)P(y=1) > P(x|y=2)P(y=2)$$

$$\Leftrightarrow \log P(x|y=1) + \log P(1) > \log P(x|y=2) + \log P(2)$$

$$(x-u_1)^T \Sigma_1^{-1} (x-u_1) - \frac{1}{2} \log |\Sigma_1| + \log P(1) > (x-u_2)^T \Sigma_2^{-1} (x-u_2) - \log |\Sigma_1| + \log P(2)$$

$$\Leftrightarrow \underbrace{x \Sigma_1^{-1} x - x \Sigma_2^{-1} x - 2x^T \Sigma_1^{-1} u_1 + 2x^T \Sigma_2^{-1} u_2}_{\text{independent of } x} + \text{constant} \geq 0$$

Decision surface for QDA where $\Sigma_1 \neq \Sigma_2$

$$\textcircled{1} \quad \underbrace{x^T (\Sigma_1^{-1} - \Sigma_2^{-1}) x - 2x^T (\Sigma_1^{-1} \mu_1 - \Sigma_2^{-1} \mu_2)}_{\text{quadratic terms}} + \text{const}$$

$\Rightarrow$ decision boundary could be quadratic.

Decision surface for LDA where $\Sigma_1 = \Sigma_2 = \Sigma$

$$- 2x^T \Sigma^{-1} (\mu_1 - \mu_2) + \text{constant} \geq 0$$

shared covariance

$\rightarrow$ linear in x .

Analyzing decision surface of Naive Bayes classifier for continuous columns.

Assume $P(x_j | y) \sim N(x_j; \mu_{jy}, \sigma_{jy}^2)$ $x_j \in \mathbb{R}$.

$$\log(P(X|Y=1)P(Y=1)) > \log(P(X|Y=2)P(Y=2))$$

$$\log\left(\prod_{j=1}^d P(x_j | y=1) P(y=1)\right)$$

$$= -\sum_{j=1}^d \left(\frac{(x_j - \mu_{j1})^2}{2\sigma_{j1}^2} - \log \sigma_{j1} \right) + \log P(y=1) > -\sum_{j=1}^d \frac{(x_j - \mu_{j2})^2}{2\sigma_{j2}^2} - \log \sigma_{j2} + \log P(y=2)$$

Training Generative Models

Training data:

$$D = \{(x_1, y_1), \dots, (x_N, y_N)\}$$

Training approach: find parameters of $P(y)$ & $P(x|y)$ so as to maximize the likelihood of data D .

Assume: θ denotes all the parameters required to estimate $P(y)$ and $P(x|y)$
we might write this as $P(y|\theta)$ & $P(x|y, \theta)$

$$\text{MLE} \quad \max_{\theta} \log P(D|\theta) = \max_{\theta} \sum_{i=1}^N \log P(\underline{x}_i, \underline{y}_i | \theta)$$

generative ~~data~~ model joint probability of (x, y) instead of just $P(y|x)$
like we did in logistic regression

$$\log P(D|\theta) = \sum_{i=1}^N \log P(y_i|\theta) P(x_i|y_i, \theta)$$

$$= \sum_{i=1}^N \log P(y_i|\theta) + \sum_{c=1}^K \left[\sum_{\substack{i: y_i=c}} \log P(x_i|y_i, \theta_c) \right]$$

$\rightarrow \sum_{i=1}^N \log P(x_i|y_i, \theta)$

We have $K+1$ probability distribution estimation tasks

- ① Estimate $P(y|\theta)$ using D
- ② Estimate for each class c , the $P(x|y, \theta_y)$
 $c \in [1 \dots K]$
↑
parameters of class c .

Estimating $P(y|\theta)$ & class priors -

$P(y|\theta) \sim$ Categorical distribution

$\sim \text{cat}(\pi_1, \pi_2, \dots, \pi_K)$

using D . $(\pi_1, \dots, \pi_K) \in \underline{\Theta}$

Also loosely referred as multinomial with # events=1

$\sum_{c=1}^K \pi_c = 1 \quad \pi_c \geq 0$

Training $\pi_1, \dots, \pi_k$ $\pi_c \equiv \text{probability of class } c$

$$\max_{\pi_1, \dots, \pi_k} \sum_{i=1}^N \log P(y_i | \pi_1, \dots, \pi_k) \equiv \max_{\pi_1, \dots, \pi_k} \sum_{i=1}^N \log \left(\prod_{c=1}^k \pi_c^{\delta(y_i=c)} \right)$$

where $\delta(b) = 1$ if $b = \text{true}$
 $= 0$ otherwise

$N_c = \#$ of examples in D with label c .

$$= \max_{\pi_1, \dots, \pi_k} \log \prod_{c=1}^k \pi_c^{N_c} \quad \text{where}$$

$$\sum_c \pi_c = 1; \pi_c \geq 0$$

$$\max_{\pi_1, \dots, \pi_k} \sum_{c=1}^k N_c \log \pi_c$$

$$\text{s.t. } \pi_c \geq 0, \sum_c \pi_c = 1$$

$\Leftrightarrow$

$$\max_{\pi_1, \dots, \pi_k} \sum_{c=1}^k N_c \log \pi_c + u(\sum_c \pi_c - 1)$$

$$\pi_c \geq 0$$

$$\nabla_{\pi_c} \left[\sum_c N_c \log \pi_c + u(\sum_c \pi_c - 1) \right] \equiv \left| \frac{N_c}{\pi_c} + u = 0 \right| \Rightarrow \pi_c = -\frac{N_c}{u}$$

$$\sum_{c=1}^K \pi_c = 1 \Rightarrow \sum_{c=1}^K \frac{N_c}{u} = 1 \Rightarrow u = -\sum_{c=1}^K N_c = -N$$

$$\Rightarrow \pi_c^{MLE} = \frac{N_c}{N}$$

$$P(y|\theta)^{MLE} \sim \text{cat}\left(\frac{N_1}{N}, \dots, \frac{N_K}{N}\right)$$

~~Training the class-specific $P(\underline{x}|y, \theta_y)$ distributions.~~

$$\textcircled{1} P(\underline{x}|y, \theta_y) \sim \mathcal{N}(\underline{x}; \mu_y, \Sigma_y)$$

estimating μ_c & Σ_c using MLE

$$\max_{\mu_c, \Sigma_c} \sum_{i: y_i = c} \log \mathcal{N}(x_i; \mu_c, \Sigma_c) \quad \leftarrow \text{covered in CS215}$$

$$- \mu_c^{\text{MLE}} = \frac{1}{N_c} \sum_{i: y_i = c} x_i$$

$$- \Sigma_c^{\text{MLE}} = \frac{1}{N_c} \sum_{i: y_i = c} (x_i - \mu_c)(x_i - \mu_c)^T$$

$$x \in \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

② LDA where Σ is shared across classes:

$$\Sigma^{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_{y_i})(x_i - \mu_{y_i})^T$$

③ Naive Bayes:

$$P(x|y, \theta_y) = \prod_{j=1}^d P(x_j|y, \theta_y)$$

MLE of this distribution:

$$\max_{\theta_c} \sum_{i: y_i = c} \sum_{j=1}^d \log P(x_{ij} | y_i, \theta_{cj})$$

parameter of j in dimension c .

3(a) Suppose: $P(x_j | y, \theta_{yj})$ if it follow a Gaussian distribution:

$$\theta_{yj} = \{ \mu_{yj}, \sigma_{yj} \}$$

$$P(x_j | y, \mu_{yj}, \sigma_{yj}) = N(x_j | \mu_{yj}, \sigma_{yj})$$

$$\mu_{yj}^{MLE} = \frac{1}{N_y} \sum_{i: y_i = y} x_{ij} \quad ; \quad \sigma_{yj}^{2, MLE} = \frac{1}{N_y} \sum_{i: y_i = y} (x_{ij} - \mu_{yj})^2$$

* j is dimension, i is example id, c or y is class label

3 (b) $P(x_j | y, \theta_{y,j}) \sim \text{Cat}(\theta_{y,j1} \dots \theta_{y,jm_j})$ where
 $m_j = \text{domain of column } j \text{ in data}$
 $x_j \in \{1, 2, \dots, m_j\}$.

$$P(x_j = v | y) \stackrel{\text{MLE}}{=} \frac{N_{y,j,v}}{N_y} \quad \# \text{ of instances in } D \text{ where } x_{ij} = v \text{ and } y_i = y$$

If $N_{y,j,v} = 0$ then $P(x_j = v | y, \theta_{y,j}) = 0$
 $\Rightarrow P(x | y, \theta) = 0$ for example with $x_j = v$
 corresponding to values of
 other column.

Bayesian Estimation of Naïve Bayes Parameters

Provides a principled mechanism of smoothing

Limitation of MLE

(Taken from my CS215 fall 2024 office)

- Over-reliance on data sample D . If data is limited, estimates can be very wrong.

- Example, Bernoulli p could be zero if no 1s in 10 trials.

- No indication on the uncertainty of the estimated parameters.

- Example, for a Bernoulli parameters whether estimation is made from two with 50% heads or 1000 examples with 50% heads, the estimated parameter is the same.

D_1 $|D_1| = 2$ $N_1(D_1) = 1$, $N - N = 1$ $\hat{p}_1 = 0.5$ ←

D_2 $|D_2| = 1000$ $N_1(D_2) = 500$ $\hat{p}_2 = 0.5$ ←

- No mechanism to specify human's prior knowledge of the parameters.

Example of limitations of MLE

- Suppose a toss a coin 10 times and get

H, H, H, H, H, H, H, H, H, H

Estimate p ? MLE: $\hat{p} = \# \text{ ones} / N = 1$

What is your guess on the probability p of head?

- Suppose you want to form a music band, and you are looking for bass guitarist. You ask 7 random batchmates: “Can you play the bass guitar” and you get answers

N, N, N, N, N, N, N D

What fraction of batchmates play bass guitar?

MLE: 0

Do you have a different guess?

$p \approx 0.01$

Bayesian estimation

- Treat the parameters as a random variable which has a distribution.
 - Step 1: Humans specify their prior knowledge of the values of the parameters as a distribution $f_{\text{pr}}(\theta)$
 - Example: $f_{\text{pr}}(\theta) \sim U(0,1)$ where θ denotes the parameter p of a Bernoulli
- Also called prior probability

Bayesian estimation

- Calculate the posterior distribution of parameters after observing data D following Bayes rule

$$f(\theta|D) = f(\theta)f(D|\theta) / \int_{\theta} f(\theta)f(D|\theta)$$

$$f_{po}(\theta|D)$$

Posterior probability

Using Bayesian estimates during deployment

$$f(\theta|D) \equiv \underbrace{f(\theta|D)}_{p_\theta}$$

- Exact Bayesian probability computation:

- Given a new $\underline{x}$, calculate $f(\underline{X}|\underline{D})$

$$\rightarrow f(x|D) = \int_{\theta} f(x|\theta) \underbrace{f(\theta|D)}_{p_\theta} d\theta$$

Binomial eg:

short cut - 1

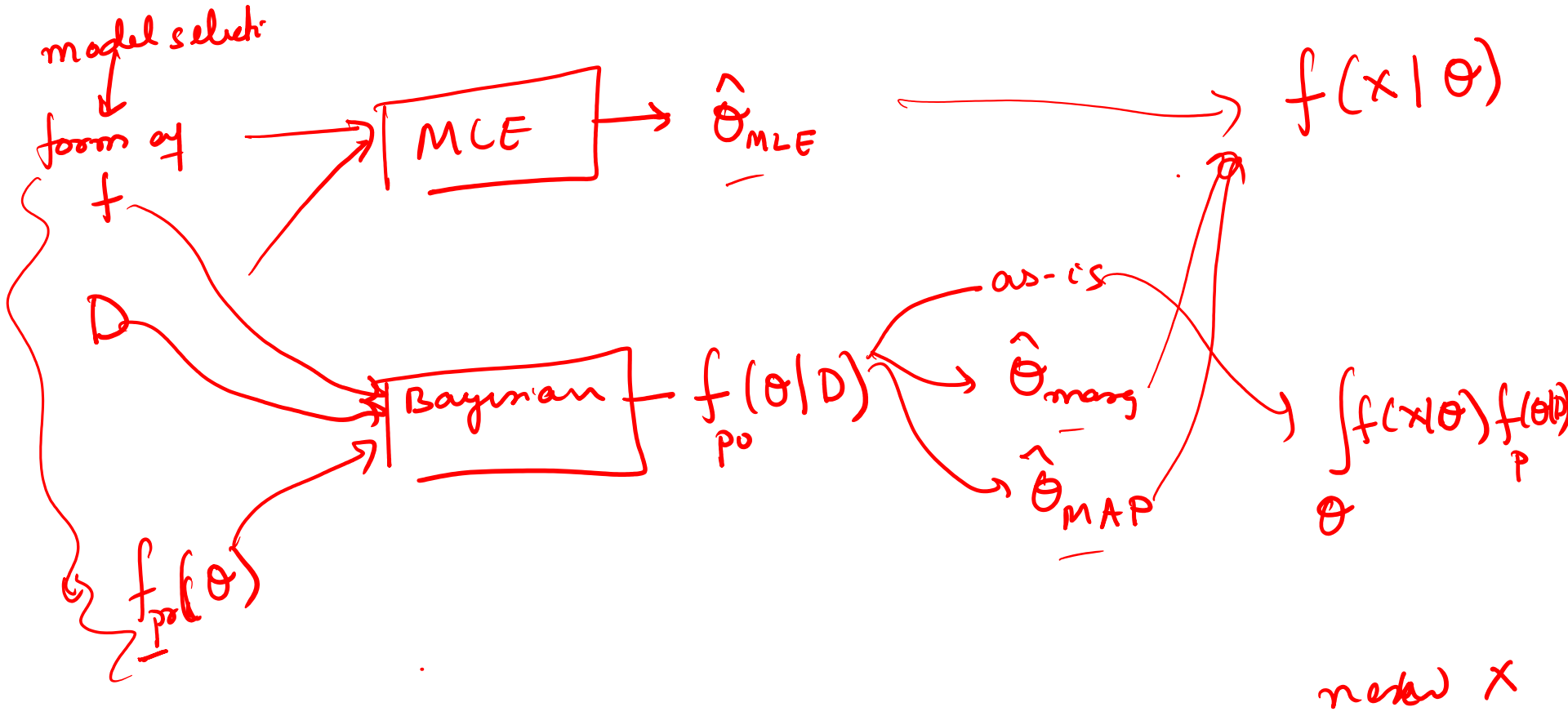
- Expected value of parameters: calculate expected value of $\underbrace{f(\theta|D)}_{p_\theta}$

$$\rightarrow \hat{\theta}_{\text{marg}} = E[\theta] = \int \theta \cdot \underbrace{f(\theta|D)}_{p_\theta} d\theta$$

- MAP estimate: use $\max_{\theta} \underbrace{f(\theta|D)}$

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f(\theta|D)$$

Overall pipeline for MLE Vs Bayesian



Example:
Bayesian estimation of
Bernoulli/Binomial parameter p

Bayesian estimation of Bernoulli parameter

$$p \in [0, \dots, 1]$$

- Choose a prior distribution over parameter θ or p of Bernoulli
 - $f_{\text{pr}}(\theta)$ $\sim U(0,1)$

- Data D has n are ones and remaining $N-n = m$ are 0s.

$$D \equiv \{x_1, x_2, \dots, x_N\} \text{ eg: } \{0, 1, 1, \dots, 0, \dots, 0\}$$

$$D \equiv \{n, m\}$$

#1s #0s.

$$f(D|\theta) = \prod_{i=1}^N f(x_i|\theta) = \theta^n (1-\theta)^{N-n} = \theta^n (1-\theta)^m$$

- Posterior distribution is:

$$\underbrace{f(\theta|D)}_{\text{Po P}} = \frac{\underbrace{f(\theta)}_{\text{Pr}} \underbrace{f(D|\theta)}}{\int_{\theta'} f(\theta') f(D|\theta')} = \frac{1 \cdot \theta^n (1-\theta)^m}{\int_{\theta'} \theta'^n (1-\theta')^m} = Z$$

$m = N-n$

$$\underline{f_{\theta}(D)} = \frac{(1-\theta)^m (\theta)^n}{Z} \quad 0 \leq \theta \leq 1$$

Z ← normalizer.

To get θ_{MAP}
mode of $f_{\theta}(D)$

$$\begin{aligned} & \max_{\theta} \frac{(1-\theta)^m \theta^n}{\theta} \\ \equiv & \frac{\partial}{\partial \theta} [(1-\theta)^m \theta^n] = -m(1-\theta)^{m-1} \theta^n + n\theta^{n-1} (1-\theta)^m = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow -m\theta + n(1-\theta) &= 0 \\ \theta(n+m) &= n \Rightarrow \theta_{MAP} = \frac{n}{m+n} \end{aligned}$$

Also calculate θ_{marg} .

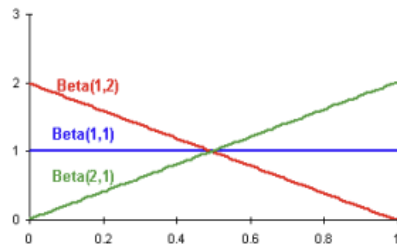
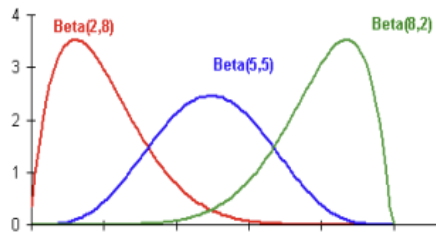
$$E[\theta] = \int_{\theta} \theta \cdot \frac{\theta^n (1-\theta)^m}{2} d\theta$$

Beta Random Variable

X is a **Beta Random Variable**: $X \sim \text{Beta}(a, b)$

- Probability Density Function (PDF): (where $a, b > 0$)

$$f(x) = \begin{cases} \frac{1}{B(a,b)} x^{a-1} (1-x)^{b-1} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

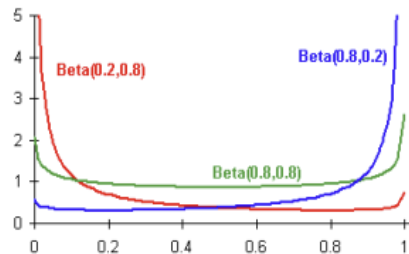


$$B(a,b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$$

$$B(z_1, z_2) = \frac{\Gamma(z_1) \Gamma(z_2)}{\Gamma(z_1 + z_2)}$$

$z_1, z_2 \equiv \text{integers}$

$$\frac{(z_1-1)! (z_2-1)!}{(z_1+z_2-1)!}$$



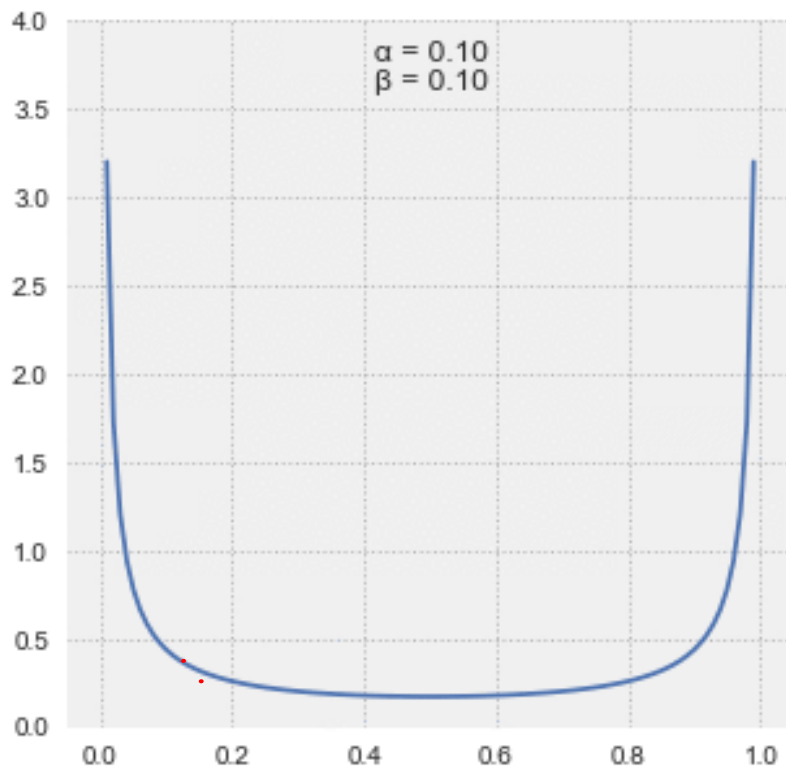
- Symmetric when $a = b$

$$E[X] = \frac{a}{a+b}$$

$$\text{Var}(X) = \frac{ab}{(a+b)^2 (a+b+1)}$$

Mode: $\frac{a-1}{a+b-2}$

The shapes of the Beta distribution



Alternative parameterization
 $P(\theta | \alpha, \beta) \triangleq$

$$f(\theta | p, s)$$

where $p = \frac{\alpha}{\alpha + \beta}$

and $s = \alpha + \beta$

Beta distribution is the
distribution of probabilities

.

More properties of Beta distributions

- Uniform distribution $U(0,1) = B(1,1)$

$$f(\theta | a, b) = \frac{\theta^{a-1} (1-\theta)^{b-1}}{B(a, b)} = \frac{1 \cdot 1}{B(1, 1)} = 1$$

- Relationship between Beta and Gamma distribution

- Let $Y = G(a, 1)$ and $W = G(b, 1)$

$$f(y | a, 1) = \frac{e^{-y} y^{a-1}}{\Gamma(a)}$$

$$f(x | \alpha, \lambda) = \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)}$$
$$f(w | b, 1) = \frac{e^{-w} w^{b-1}}{\Gamma(b)}$$

- The $X = Y/(Y+W)$ follows a Beta distribution $B(a, b)$

$$X = \frac{Y}{Y+W} \quad \text{then} \quad X \sim B(a, b)$$

Expected value of the posterior of Binomial

$$f_{p_0}(\theta | D) \equiv \frac{\theta^n (1-\theta)^m}{2} \equiv B(a=n+1, b=m+1)$$

$$D \equiv \begin{matrix} \{n, m\} \\ \uparrow \quad \uparrow \\ \#1s \quad \#0s \end{matrix}$$

$$\hat{\theta}_{\text{marg}} = \frac{n+1}{n+m+2}$$

Laplace smoothing..

Contrast with MLE

$$\hat{\theta}_{\text{MLE}} \equiv \frac{n}{m+n}$$

$$\rightarrow f_{\text{prior}}(\theta) = \text{Beta}(p=0.1, S=10)$$

$$\equiv \text{Beta}(1, 9) \equiv \text{Beta}(a_0, b_0)$$

D: Dataset: n ones and $N-n$ zeros

$$f_{\text{po}}(\theta|D) \propto \underbrace{\theta^{a_0-1} (1-\theta)^{b_0-1}}_{f_{\text{pr}}(\theta)} \cdot \underbrace{\theta^n (1-\theta)^{N-n}}_{f(D|\theta)}$$

$$= \theta^{n+a_0-1} (1-\theta)^{N-n+b_0-1} = \text{Beta}(n+a_0, N-n+b_0)$$

Conjugate prior

- Assume prior distribution is: $\theta \sim f_{pr}(\theta) = B(a, b)$
- Assume dataset D has n are ones and remaining $N-n = m$ are 0s.
- Posterior:
-

Conjugate priors

- A distribution $f(X; \theta)$ has a conjugate prior $f_{pr}(\theta)$ if the posterior distribution $f(\theta|D)$ has the same function form as the prior distribution

- Example:

$$f_{prior}(\theta) = \text{Beta}(p, q) = \frac{\theta^{p-1} (1-\theta)^{q-1}}{B(p, q)}$$

$D = \{m \text{ successes}, n-1 \text{ trials}\}$

$$f_{post}(\theta|D) = \frac{\theta^{p-1} (1-\theta)^{q-1} [\theta^n (1-\theta)^m]}{B(p, q)} = \frac{\theta^{n+p-1} (1-\theta)^{m+q-1}}{B(n+p, m+q)}$$

$$f_{prior}(\theta) = \frac{\text{Beta}(50, 50)}{\text{Beta}(1, 1)} \theta \propto \text{Beta}(n+p, m+q)$$

$\text{Beta}(5000, 5000)$

Effect of prior reduces with increasing N

Example:
Bayesian estimation of
Multinomial parameter

$$X \in \{1, 2, \dots, m\}$$

$$X \sim \text{cat}(\pi_1, \pi_2, \dots, \pi_m)$$

Given data: $D = \{x^1, x^2, \dots, x^N\}$ Counts of each out come

$$x^i \in \{1, \dots, m\} \equiv \{N_1, N_2, \dots, N_m\}$$

$$f_{\text{pr}}(\pi) \equiv 1 \quad \forall \quad \pi \in \Delta^{m-1}$$

space of all m dimensional vectors which look like multinomial probabilities

$$f_{\text{po}}(\pi | D) = \frac{f_{\text{pr}}(\pi) f(D | \pi)}{Z}$$

$$= 1 \cdot \prod_{j=1}^m \pi_j^{N_j}$$

$$= \frac{\prod_{j=1}^m \pi_j^{N_j}}{Z}$$



Dirichlet distribution
with parameters
 $(N_1+1, \dots, N_m+1)$

Dirichlet distribution generalizes
Beta distribution for
multivariate categories.

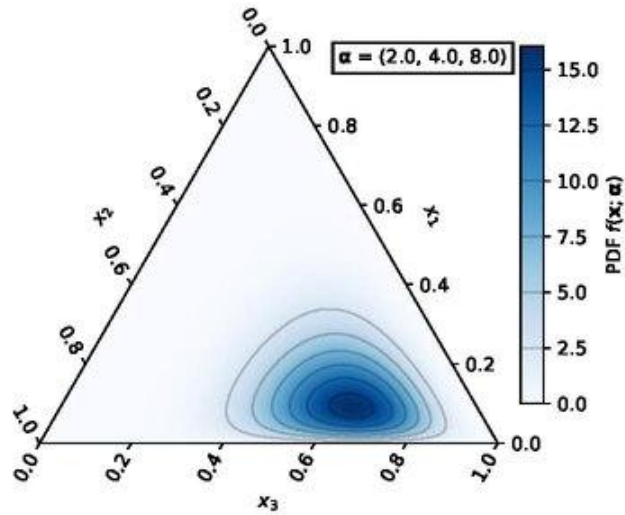
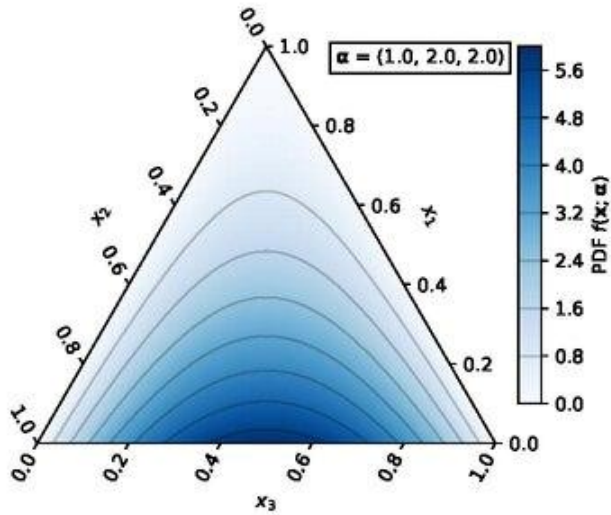
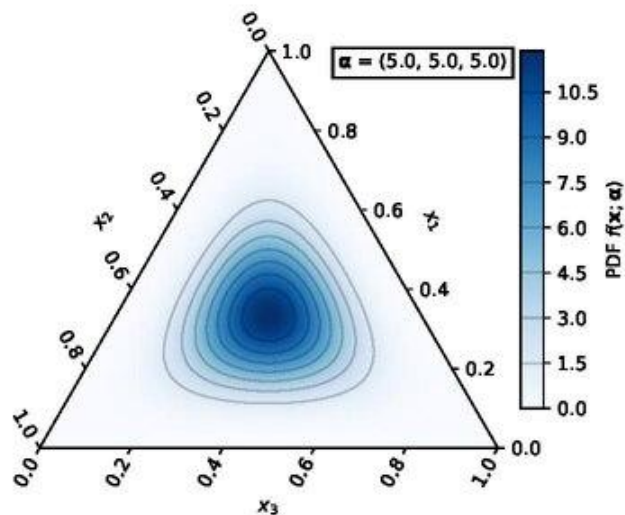
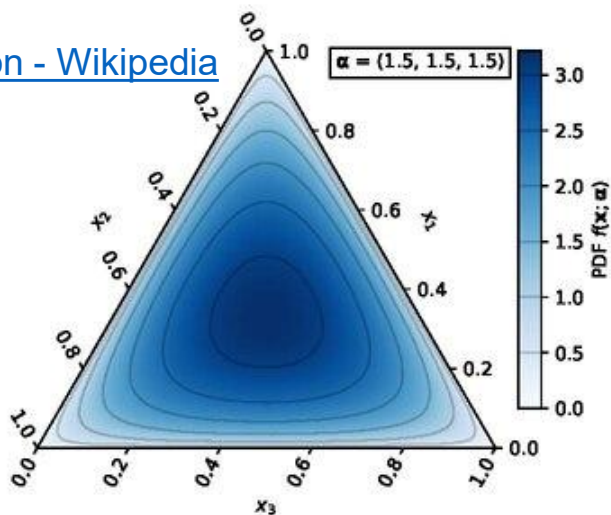
In general if $\pi_1, \dots, \pi_m \sim \text{Dir}(\alpha_1, \dots, \alpha_m)$

$$f(\pi_1, \dots, \pi_m) = \frac{\prod_{j=1}^m \pi_j^{\alpha_j - 1}}{B(\alpha_1, \dots, \alpha_m)} \quad \text{if } \begin{cases} \alpha_j \geq 0 \\ \pi_j \geq 0 \\ \sum_j \pi_j = 1 \end{cases}$$

where $B(\alpha_1, \dots, \alpha_m)$

$$= \frac{\prod_{j=1}^m \Gamma(\alpha_j)}{\Gamma(\alpha_1 + \alpha_2 + \dots + \alpha_m)}$$

Gamma function



Dirichlet distribution

$$E(\pi_j) = \frac{\alpha_j}{\underbrace{\alpha_1 + \dots + \alpha_m}_{= \alpha}}$$

$$\text{mode of } \pi_j = \frac{\alpha_j - 1}{\alpha - m}$$

* Show that Dirichlet distribution is conjugate prior of categorical / multinomial distribution

* Other conjugate priors:

Gaussian ^{over mean} is conjugate of Gaussian data.